

Paper 2 - Set 15 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Exponentials and logarithms
2	C	Trigonometric functions and equations
3	D	Integration
4	D	Mathematical proof
5	F	Angles, polygons, congruence and vectors
6	D	Mathematical proof
7	A	Sequences and series
8	F	Chains of reasoning
9	C	The logic of arguments
10	A	Identifying errors in proofs
11	H	Straight lines
12	B	The logic of arguments
13	B	The logic of arguments
14	F	Functions and transformations
15	H	Indices and surds
16	H	Binomial expansion
17	A	Identifying errors in proofs
18	F	Functions and transformations
19	C	Chains of reasoning
20	F	Chains of reasoning

Worked solutions

Question 1 LOG-11 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

Answer: C

$$\frac{3}{2}\log_2 x = 6 \Rightarrow \log_2 x = 4 \Rightarrow x = 16.$$

Question 2 TRG-01 · Trigonometric functions and equations · spec MM4.6

How many solutions does $\sin 2x = \frac{1}{2}$ have for $0 \leq x < 2\pi$?

Answer: C

$2x$ runs over $[0, 4\pi)$, which contains two full periods of $\sin$, each giving two solutions: 4 in total.

Question 3 INT-16 · Integration · spec MM7.4

Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

Answer: D

$$10 + 24 - (-10) = 44.$$

Question 4 PRF-02 · Mathematical proof · spec Prf1

Which of the following is a counterexample to the claim '**if n is a positive integer, then $2^n + 1$ is prime**'?

Answer: D

$2^3 + 1 = 9 = 3 \times 3$ is not prime. ($n = 1, 2, 4, 8$ give 3, 5, 17, 257, all prime.)

Question 5 GEO-10 · Angles, polygons, congruence and vectors · spec M5.3

Which of the following statements is/are true?

- I Every rhombus is a parallelogram.
- II Every rectangle is a rhombus.
- III The diagonals of a kite are perpendicular.

Answer: F

A rhombus has two pairs of parallel sides (I). A rectangle need not have equal sides (II false). The diagonals of a kite cross at right angles (III).

Question 6 PRF-08 · Mathematical proof · spec Prf2

It is given that for all real x : if $f(x) > 0$ then $g(x) > 0$. It is also known that $g(2) = -1$. What can be deduced?

Answer: D

Contrapositive: if $g(x) \leq 0$ then $f(x) \leq 0$. Since $g(2) = -1 \leq 0$, $f(2) \leq 0$ (it could be 0 or negative).

Question 7 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Question 8 RSN-16 · Chains of reasoning · spec Prf5

The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

Answer: F

$(m - 4) + (m - 2) + m + (m + 2) + (m + 4) = 5m$ where m is the middle (odd) number; $5m$ is odd, so 10 is never a factor.

Question 9 LGC-07 · The logic of arguments · spec Arg2

Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?

Answer: C

A square has four equal sides ($P \Rightarrow Q$, sufficient), but a rhombus that is not a square shows Q does not imply P (not necessary).

Question 10 X2-25 · Identifying errors in proofs · spec Err2

A student solves $x^2 - 5x + 6 > 0$ as follows.

I $x^2 - 5x + 6 = (x - 2)(x - 3)$.

II So $(x - 2)(x - 3) > 0$.

III So $x - 2 > 0$ and $x - 3 > 0$.

IV So $x > 3$.

Which of the following is correct?

Answer: A

The full solution set is $x < 2$ or $x > 3$.

Question 11 X2-05 · Straight lines · spec MM3.1

Two distinct lines have equations $y = m_1x + c_1$ and $y = m_2x + c_2$. Which of the following statements is/are true?

I If $m_1m_2 = -1$ then the lines meet at exactly one point.

II If $m_1 = m_2$ then the lines do not meet.

III If the lines meet on the y -axis then $c_1 = c_2$.

Answer: H

I: perpendicular lines are not parallel, so they meet once. II: equal gradients and distinct lines means parallel and different, so no intersection. III: on the y -axis $x = 0$, so $y = c_1 = c_2$.

Question 12 LGC-23 · The logic of arguments · spec Arg1

Consider the statement: **if a and b are irrational, then $a + b$ is irrational.** Which of the following pairs is a counterexample?

Answer: B

A counterexample needs the hypothesis true (both irrational) and the conclusion false ($a + b$ rational):

$$\sqrt{2} + (-\sqrt{2}) = 0.$$

Question 13 LGC-05 · The logic of arguments · spec Arg3

Which of the following statements is/are true?

- I For every real number x , there exists a real number y such that $y > x$.
- II There exists a real number y such that for every real number x , $y > x$.
- III For every real number x , there exists a real number y such that $xy = 1$.

Answer: B

I is true (take $y = x + 1$). II claims a single largest real number exists: false. III fails for $x = 0$.

Question 14 FUN-12 · Functions and transformations · spec MM1.7

Let $f(x) = \sqrt{x-3}$ for $x \geq 3$ and $g(x) = x^2 + 3$. Which of the following statements is/are true?

- I $g(f(x)) = x$ for all $x \geq 3$
- II $f(g(x)) = x$ for all real x
- III $f(g(x)) = |x|$ for all real x

Answer: F

I: $g(f(x)) = (\sqrt{x-3})^2 + 3 = x$ for $x \geq 3$: true. $f(g(x)) = \sqrt{x^2 + 3} = |x|$, so II fails (e.g. $x = -1$) and III is true.

Question 15 X2-12 · Indices and surds · spec MM1.1

a and b are positive real numbers. Which of the following statements is/are true?

- I If $a^2 = b^2$ then $a = b$.
- II If $2^a = 2^b$ then $a = b$.
- III If $a^b = 1$ then $a = 1$.

Answer: H

I holds because $a, b > 0$ rules out $a = -b$. II holds because 2^x is one-to-one. III: for positive a and b , $a^b = 1$ gives $b \log a = 0$, so $\log a = 0$ and $a = 1$. All three are true.

Question 16 BIN-12 · Binomial expansion · spec MM2.4

The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?

- I $k = 0$
- II $k = 3$
- III $k = -6$

Answer: H

General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$ has power $9 - r - 2r = 9 - 3r$, which takes values $9, 6, 3, 0, -3, \dots, -18$. $k = 0$ needs $r = 3$: yes. $k = 3$ needs $r = 2$: yes. $k = -6$ needs $r = 5$: yes. All three.

Question 17 ERR-13 · Identifying errors in proofs · spec Err1

Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

- I Suppose $x + \frac{1}{x} \geq 2$.
- II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.
- III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.
- IV Hence the claim is proved.

Which of the following best describes the argument?

Answer: A

Starting from the conclusion and reaching a truth does not prove the conclusion unless every step is reversible and the argument is presented in the forward direction. Here the steps are reversible, so the repair is to write it backwards.

Question 18 FUN-10 · Functions and transformations · spec MM8.2

$f(x) = x^2 + 1$. Which of the following statements is/are true?

- I $f(f(x)) = x^4 + 2x^2 + 2$ for all x
- II $f(x + 1) = f(x) + 1$ for all x
- III $f(2x) = 4f(x) - 3$ for all x

Answer: F

I: $f(f(x)) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$; true. II: $f(x + 1) = x^2 + 2x + 2 \neq x^2 + 2$; false. III: $f(2x) = 4x^2 + 1$ and $4f(x) - 3 = 4x^2 + 1$; true.

Question 19 RSN-06 · Chains of reasoning · spec Prf5

Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

Answer: C

36 makes all three statements true, not exactly two. Each of the others makes exactly two true: 16 and 25 (square, < 50), 48 and 12 (multiple of 6, < 50), 144 (square, multiple of 6).

Question 20 RSN-18 · Chains of reasoning · spec Prf5

How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

Answer: F

$4n + 4m = mn \Rightarrow (m - 4)(n - 4) = 16$. Positive factor pairs of 16 as ordered pairs: (1, 16), (2, 8), (4, 4), (8, 2), (16, 1): five. (Negative factor pairs would make m or n non-positive.)